

SDST 3612

Final project :

Brain tumor classification

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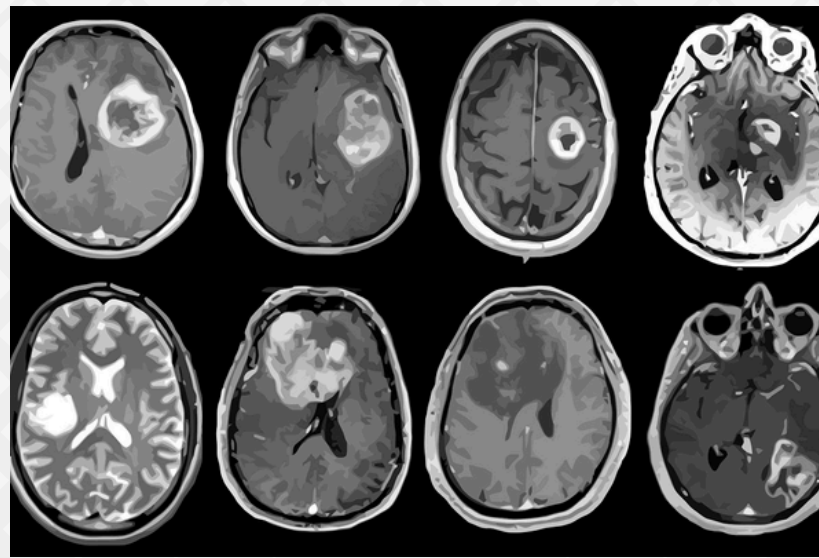
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Introduction



=> identify the tumor type as accurately as possible, because different tumors require very different treatments and have different prognoses



However, different tumor types often share very similar visual characteristics

=> **Machine Learning**



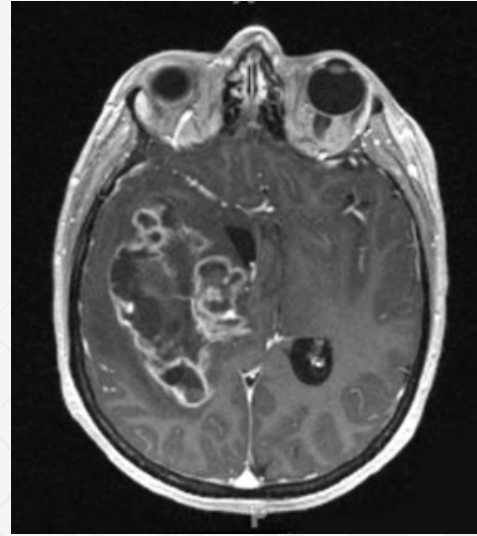
Multimodal dataset with :

- deep image features extracted from MRI
- radiomics features
- clinical and demographic data
- radiology reports

- Glioma
- Meningioma
- Brain metastasis
- Pineal / choroid plexus tumors
- Tumors of the sellar region

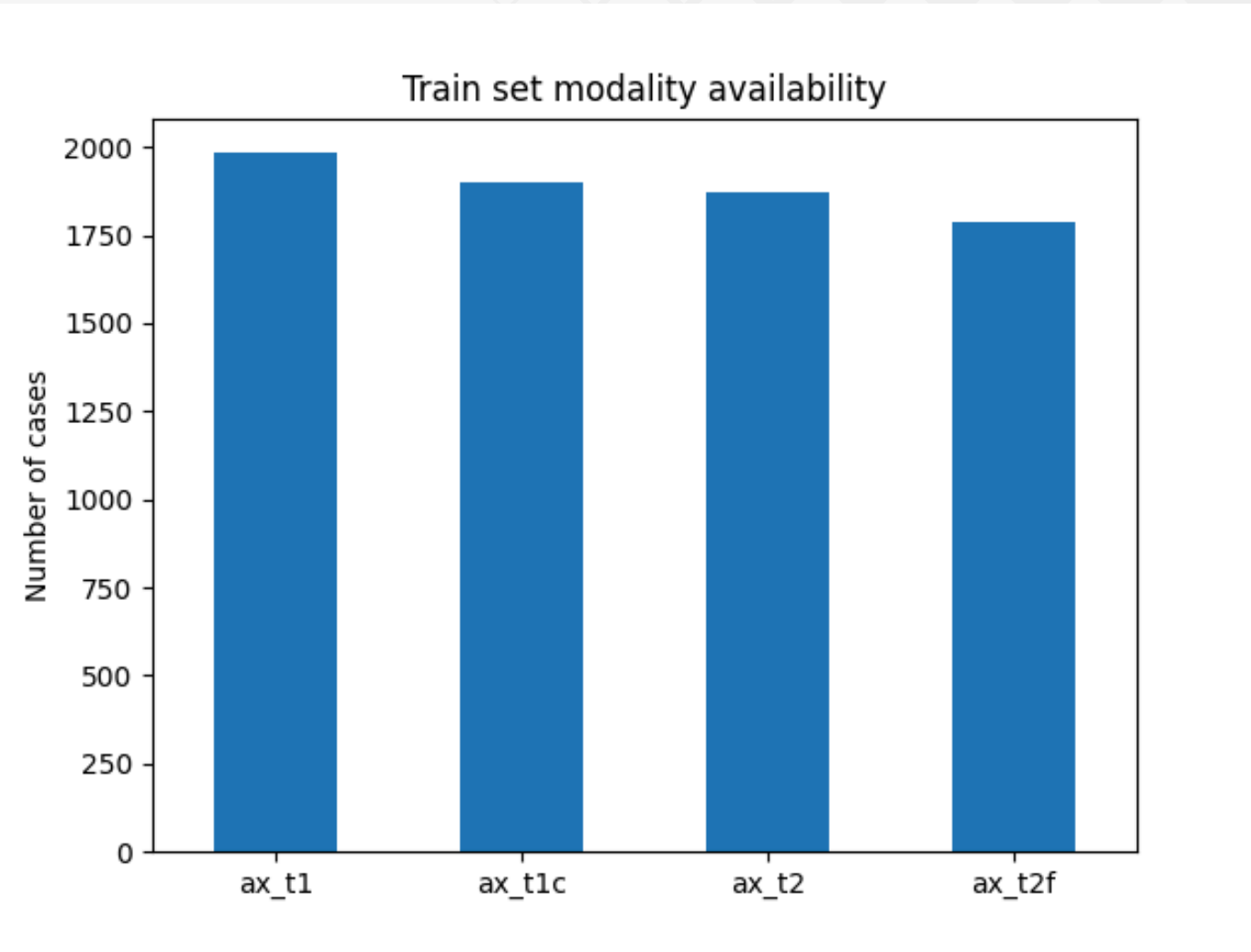
=> dataset contains about **2,800 patients**, split into training, validation, and test sets

Image features modality

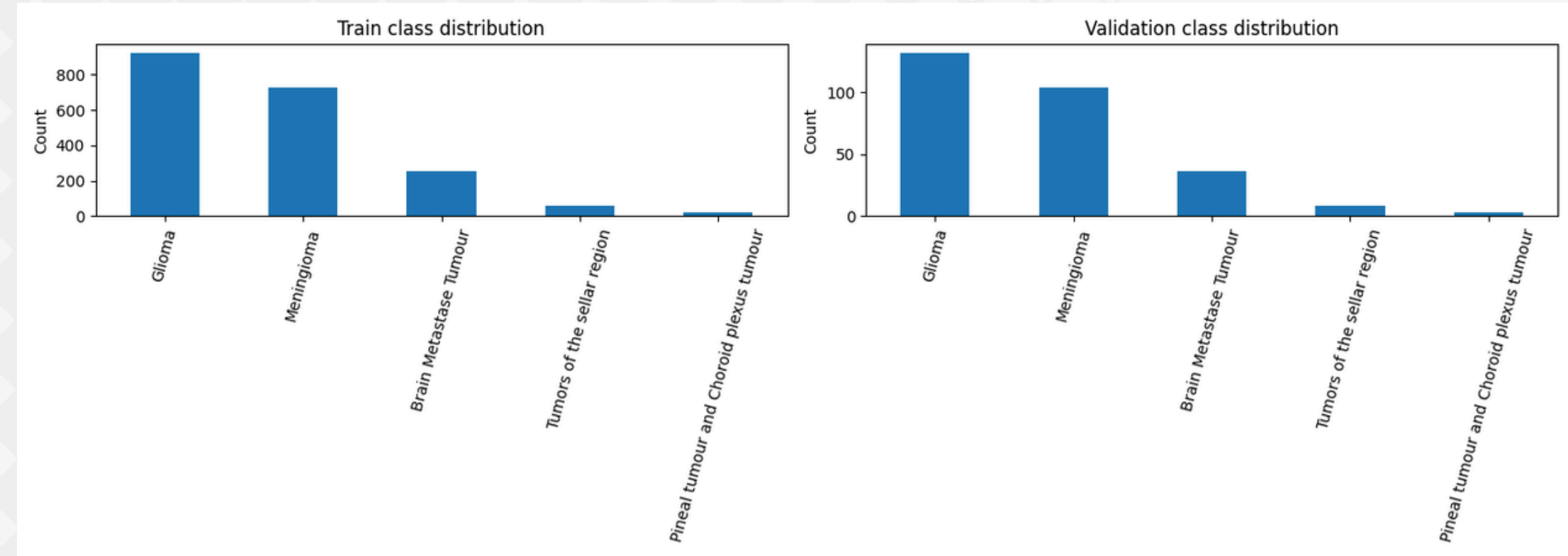


=> deep features extracted from MRI scans using a pretrained ResNet model
Each MRI modality — **T1, T1c, T2, and T2-FLAIR** — is represented by a 2048-dimensional vector => encodes **high-level visual information from the tumor**

Availability of each MRI modality



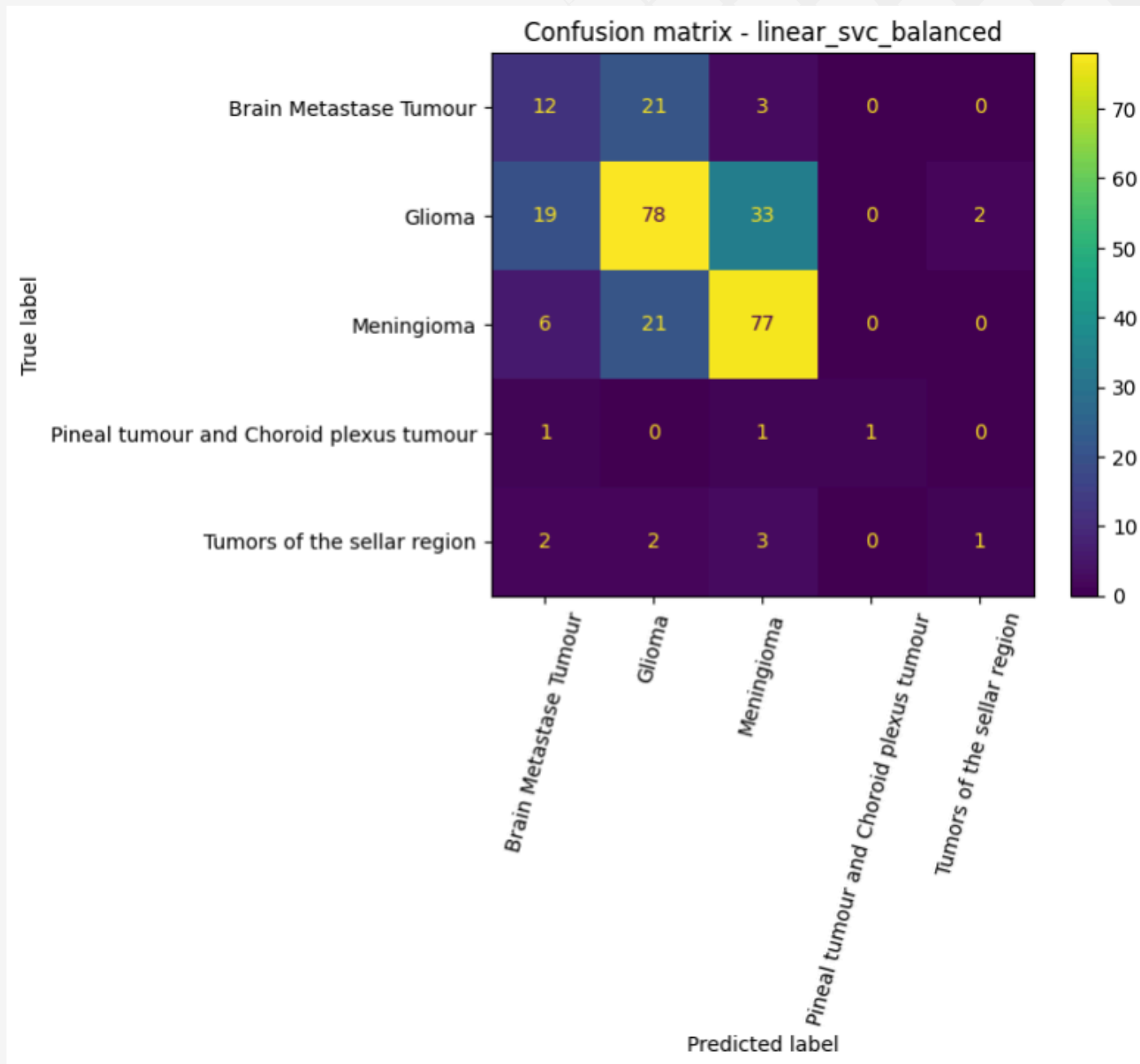
Class distribution plot



strong class imbalance in the dataset => **Glioma** and **Meningioma** dominate while some tumor types have very few samples.

Image features modality

=> best model : **linear SVC**, micro-F1 : 0.5972
model performs well on Glioma and Meningioma
(majority classes) but it struggles on rare classes



class-wise performance of Linear SVC

Class	Precision	Recall	F1-score
Brain Metastase Tumour	0.3000	0.3333	0.3158
Glioma	0.6393	0.5909	0.6142
Meningioma	0.6581	0.7404	0.6968
Pineal tumour and Choroid plexus tumour	1.0000	0.3333	0.5000
Tumors of the sellar region	0.3333	0.1250	0.1818

Table 2: Class-wise performance of the best unimodal image-features model (Linear SVC).



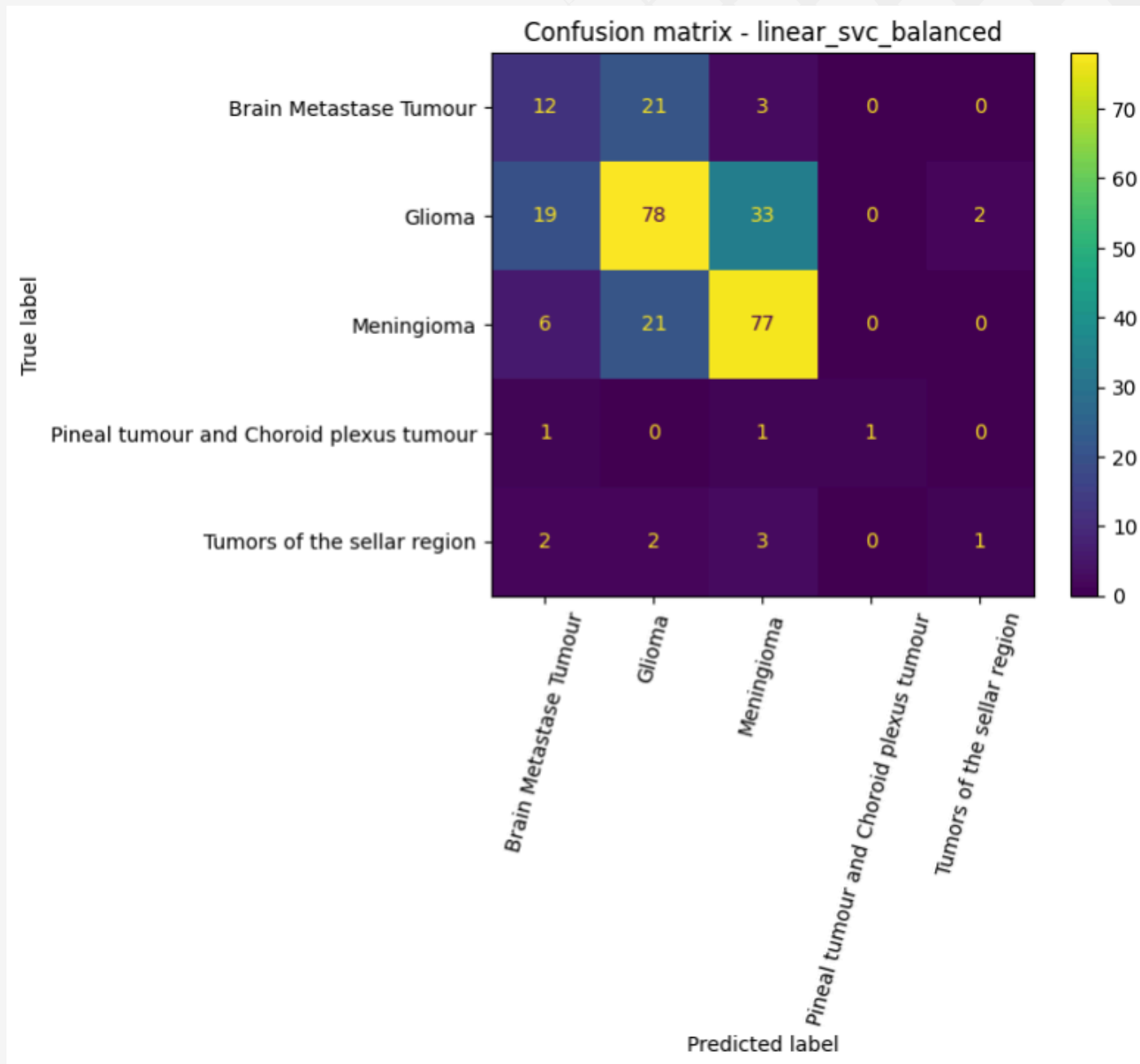
Interpretation : From a medical perspective, **low recall** on rare tumor types is **problematic**, because missing a tumor class can have **serious consequences**

=> **impact of class imbalance**

Image features alone are **not sufficient** for all classes
=> need **multimodal fusion** to improve performance

Radiomics

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Multiple-Model Training

model	f1_macro	f1_weighted	f1_micro
random_forest_balanced	0.219987	0.448209	0.522876
extra_trees_balanced	0.192911	0.403909	0.483660
xgb_balanced	0.189461	0.392815	0.450980
logreg_balanced	0.183650	0.255886	0.235294
linear_svc_balanced	0.176985	0.323804	0.294118



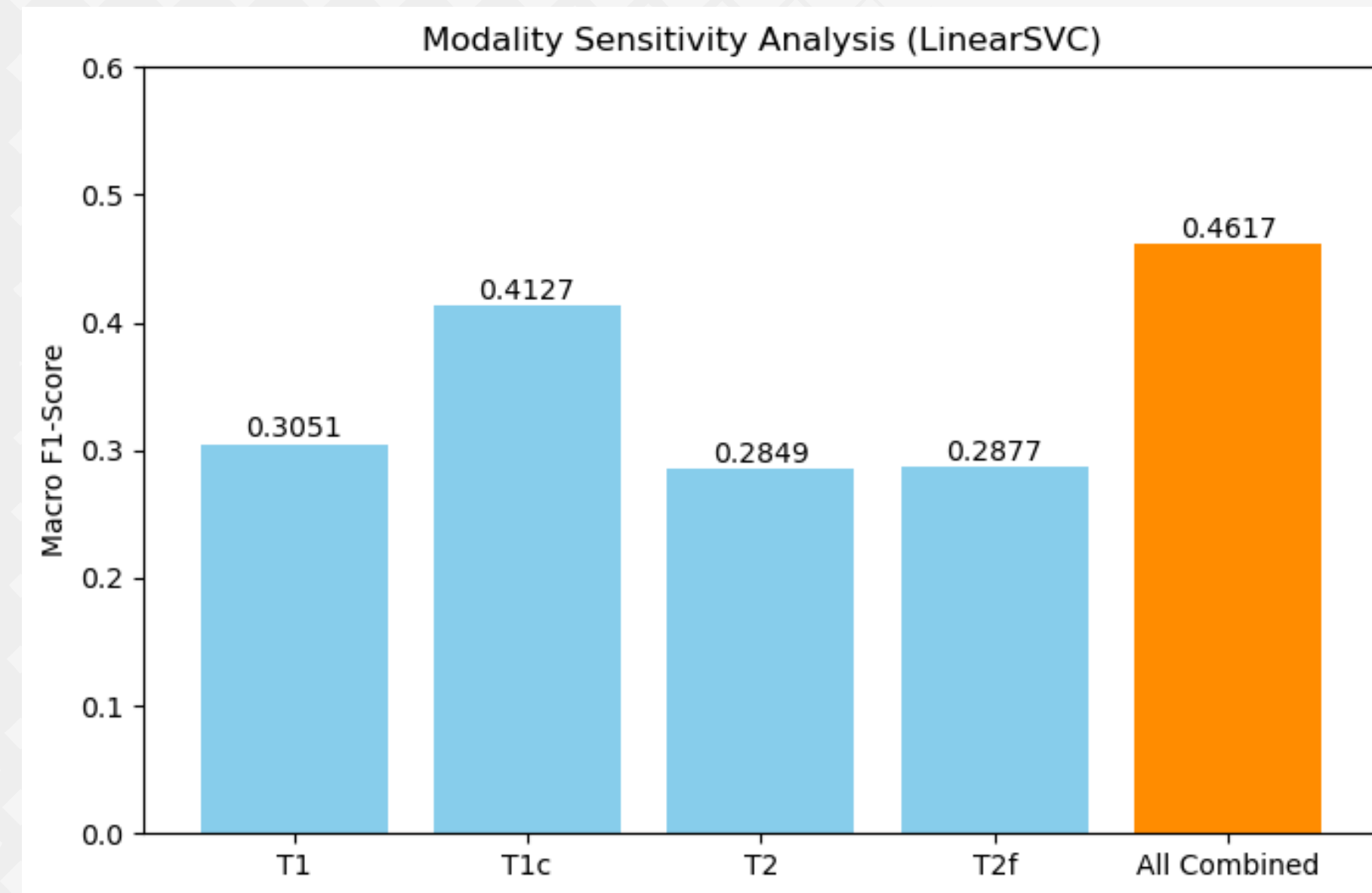
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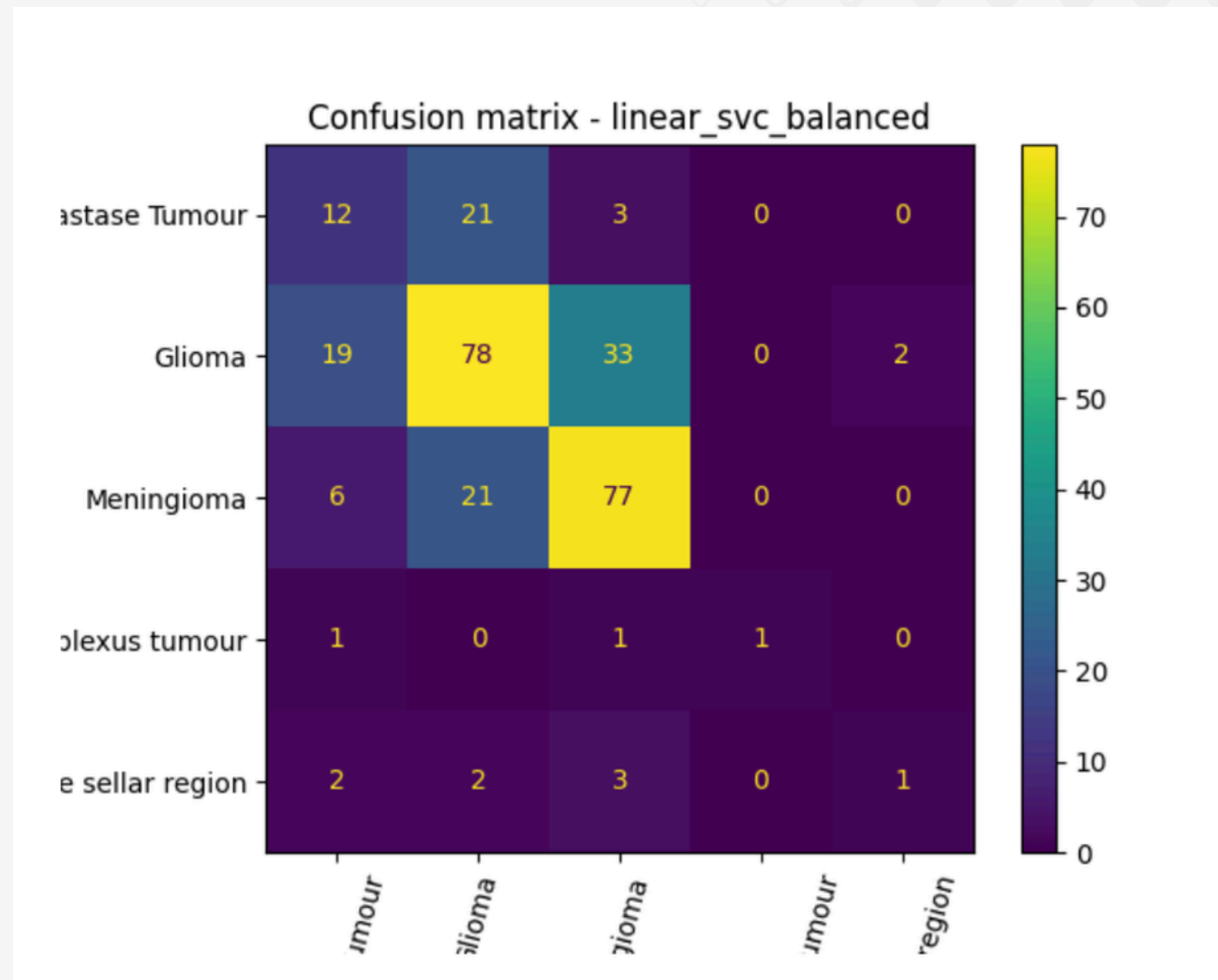
Image features sensitivity analysis (ablation study)

- **Motivation:** The combined image feature vector is massive (8,192 dimensions). Is this all useful signal, or is it mostly noise?
- **Method (Ablation Study)**
 - a. We isolated each of the 4 MRI modalities (T1, T1c, T2, T2f).
 - b. We trained the baseline Linear SVC model on each one individually.
 - c. We compared their performance to the full model.

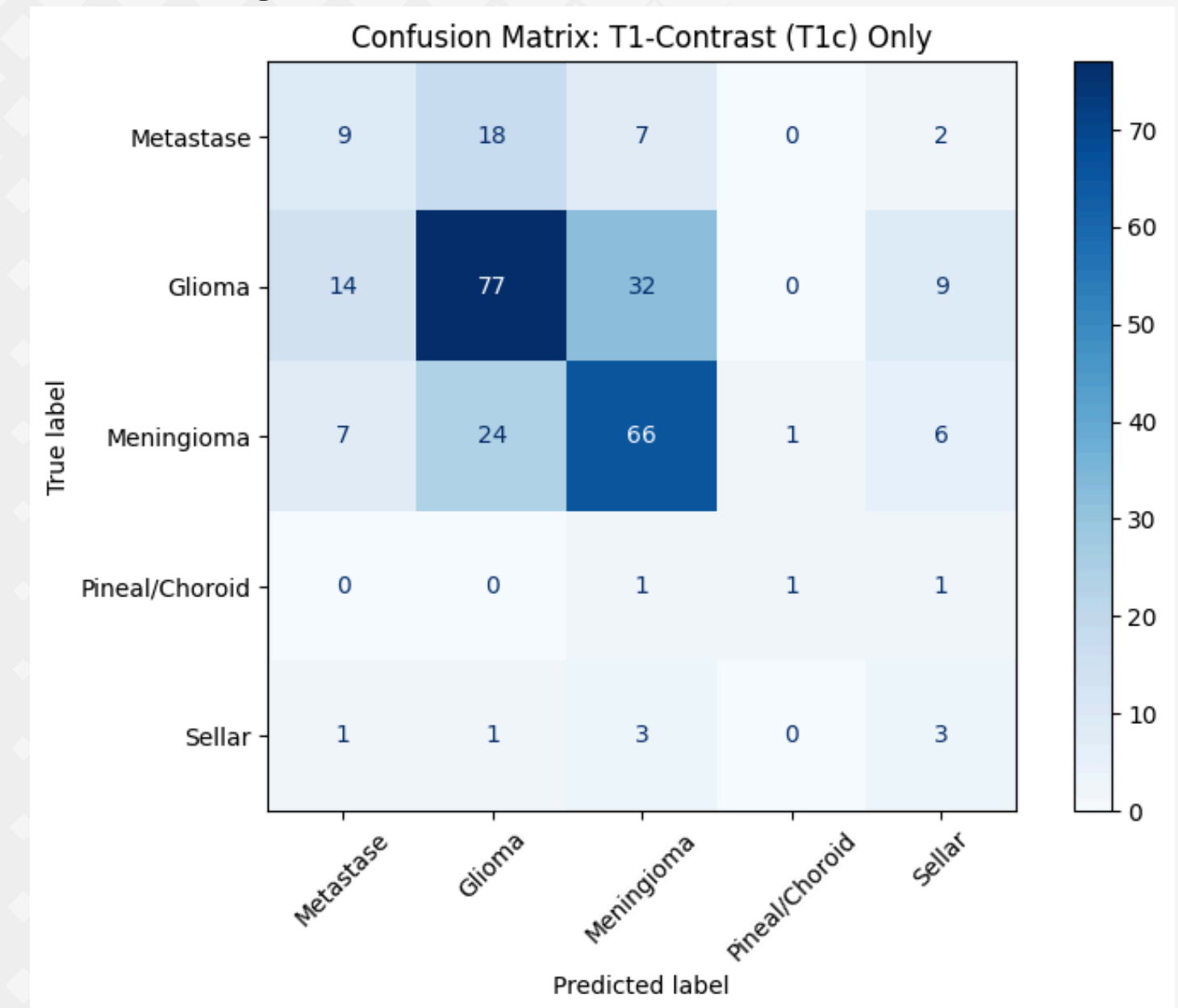


Key Insight: High Redundancy in Image Features

Full Model



T1c only



Adding 6,000+ features from T1, T2, and T2f did not fix the core confusions ! (e.g., Glioma vs. Meningioma).

A photograph of two surgeons in an operating room, wearing blue scrubs, masks, and caps, illuminated by bright surgical lights. The text "Clinical Information" is overlaid in the center in a bold, orange font.

Clinical Information

Research Objective & Data Preprocessing

Research Objective

Extract interpretable, predictive signals from unstructured radiology reports and clinical metadata, to build a robust tabular modality model for multimodal intracranial tumor classification.

Model Design Target

Aim for high interpretability while maintaining competitive predictive accuracy across multiple tumor subtypes.

Step 1: Anti-leakage Data Preprocessing

Categorical Variables Encoding

Applied Label Encoding to features: Gender, Tumor Location, and MRI sequence signal intensity levels.

Continuous Variable Imputation

Performed Median Imputation for missing age values, followed by standard Z-score normalization.

Strict Leakage Control Protocol

All feature scaling is fitted only on the training set of each CV fold to eliminate data snooping bias.

Step 2: Dual-Pathway Text Feature Extraction

Design Logic: Complementary Feature Integration

Combining statistical learning algorithms with domain-specific clinical prior knowledge to systematically convert unstructured free-text radiology reports into high-dimensional, modellable quantitative features.

Pathway 1: TF-IDF Statistics

Numerical Transformation:

Convert raw text tokens into numerical vectors using the TF-IDF weighting scheme.

Dimensionality Control:

Select the **Top 1000** most informative features to optimize the balance between semantic completeness and computational efficiency for downstream modeling.

Pathway 2: Knowledge-Driven Rules

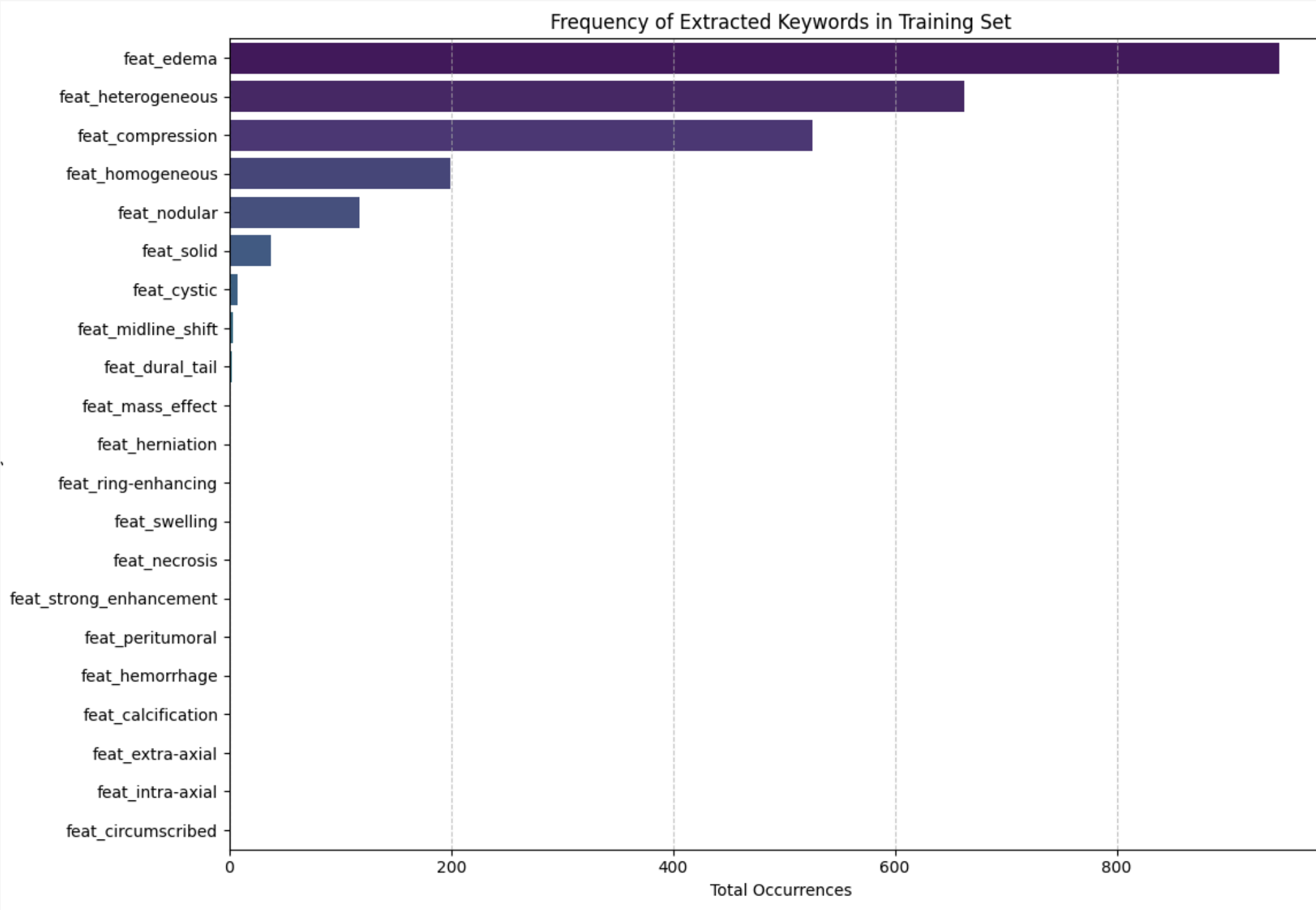
Targeted Extraction:

A rule-matching framework to identify **21 key imaging features** specific to intracranial tumors.

3 Core Categories:

1. Mass effect-related / 2. Enhancement pattern-related / 3. Tumor internal structure-related.

Step 3.1: Feature Frequency Distribution Analysis



Analysis Action & Purpose

Count the occurrence frequency of 21 extracted keyword features in the training set, to evaluate feature validity, distribution balance, and provide a reliable reference basis for subsequent feature selection strategies.

Intermediate Conclusion

Significant Imbalance: **Edema (>900)** is dominant, followed by heterogeneity, compression, homogeneity (all > 200 occurrences).

Low-Frequency Features: **Midline shift, dural tail** signand necrosis show near-zero frequency, indicating limited discriminative power.

Step 3.2: Feature Distribution Across Tumor Categories



Visualization Insight

A heatmap visualization quantifies the proportion of each keyword feature across five major intracranial tumor subtypes (Meningioma, Glioma, Pineal, Choroid Plexus, etc.). It provides a clear visual representation of feature variability and highlights subtype-specific enrichment patterns.

Core Intermediate Conclusion

Subtype-Specific:

Edema (87% in Pineal/ Choroid vs 23% in Meningioma); Homogeneity (27% only in Meningioma).

Key Takeaway:

Confirms extracted features have preliminary classification predictive value.

Calculate keyword proportions across 5 intracranial tumor subtypes to verify the classification discriminative power of text features.

Step 3.3: Statistical Significance Validation

	Feature	Chi2_Stat	p_value	Significant (p<0.05)
3	feat_homogeneous	352.131524	6.077790e-75	True
2	feat_edema	319.912872	5.476599e-68	True
1	feat_compression	52.781653	9.467054e-11	True
4	feat_heterogeneous	32.180322	1.757430e-06	True
7	feat_solid	14.855056	5.011477e-03	True
5	feat_nodular	14.823141	5.082449e-03	True
6	feat_cystic	8.051148	8.972245e-02	False
8	feat_dural_tail	3.451283	4.853244e-01	False
0	feat_midline_shift	3.443521	4.865179e-01	False

Validation

Perform chi-square test between each keyword feature and tumor classification labels, with a significance threshold of $p < 0.05$, to filter statistically robust predictive features.

Core Filtering Result

6 Features Retained:

Signal homogeneity, edema, compression, heterogeneity, solid component, nodularity.

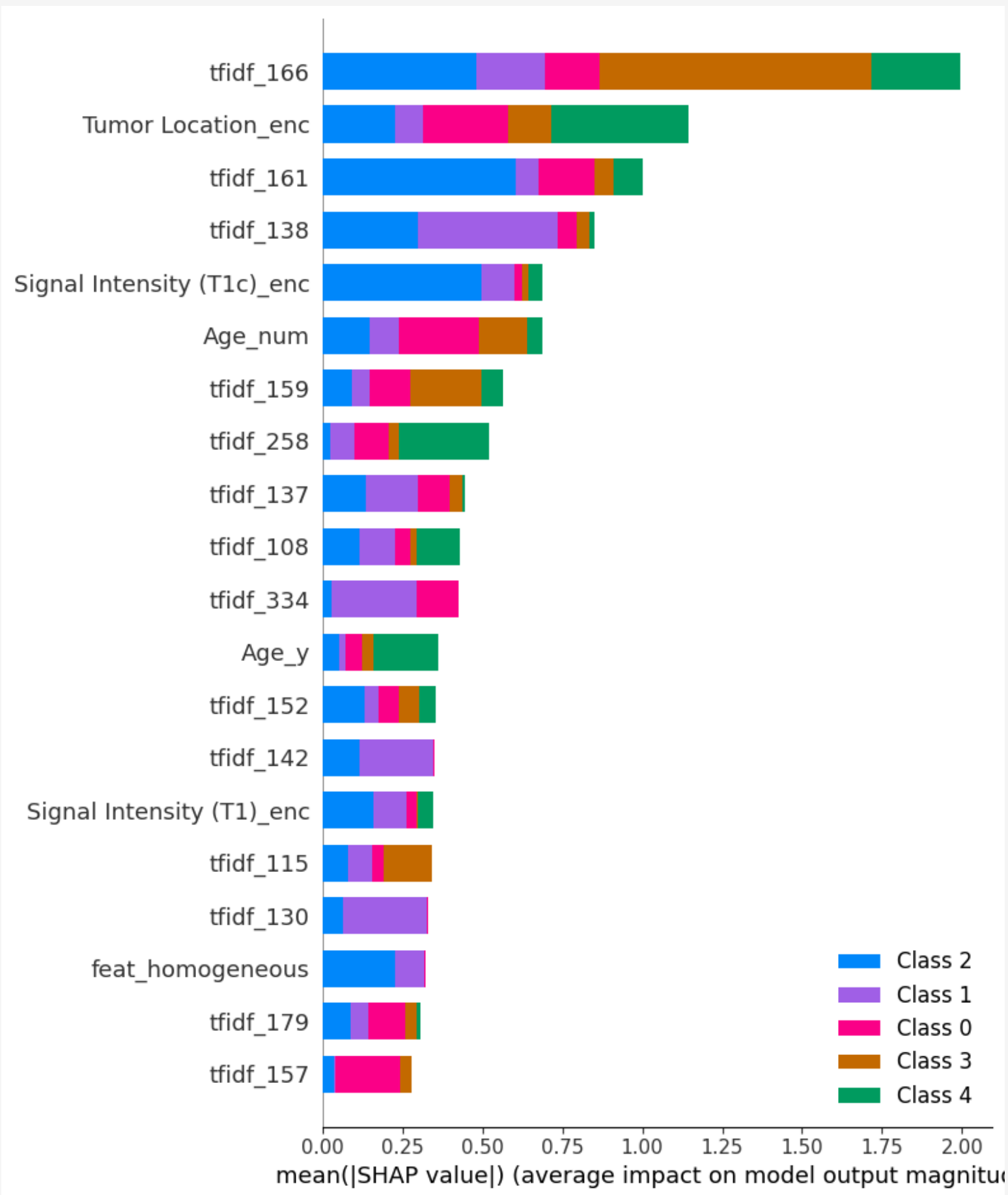
Extremely Strong:

Signal homogeneity & edema (p-values < 1e-67).

3 Excluded:

Cystic degeneration, dural tail sign, midline shift.

Feature Optimization, Model Building & Interpretability



Step 4: Feature Optimization & Dataset Refinement

Variance Threshold: Filter out low-impact features (**Variance** < 0.01).
Collinearity Filtering: Remove redundant features with **Correlation** > 0.85.
Final Dataset: Merges clinical metadata, validated keywords & TF-IDF vectors.

Step 5: Heterogeneous Ensemble Model Training

Architecture: LightGBM + RF + XGBoost, hyper-tuned with Optuna.
Performance: Achieved **0.74 average F1-score** on 5-fold cross-validation.

Step 6: Interpretability & Multimodal Fusion

SHAP Analysis: Verified model decisions align with clinical diagnostic logic.
Fusion Prep: PCA-reduced TF-IDF outputs soft probabilities for imaging fusion.

Model solely trained by Clinical Information

Fold 1 Micro F1: 0.7181

Fold 2

Fold 2 Micro F1: 0.7439

Fold 3

Fold 3 Micro F1: 0.7528

Fold 4

Fold 4 Micro F1: 0.7263

Fold 5

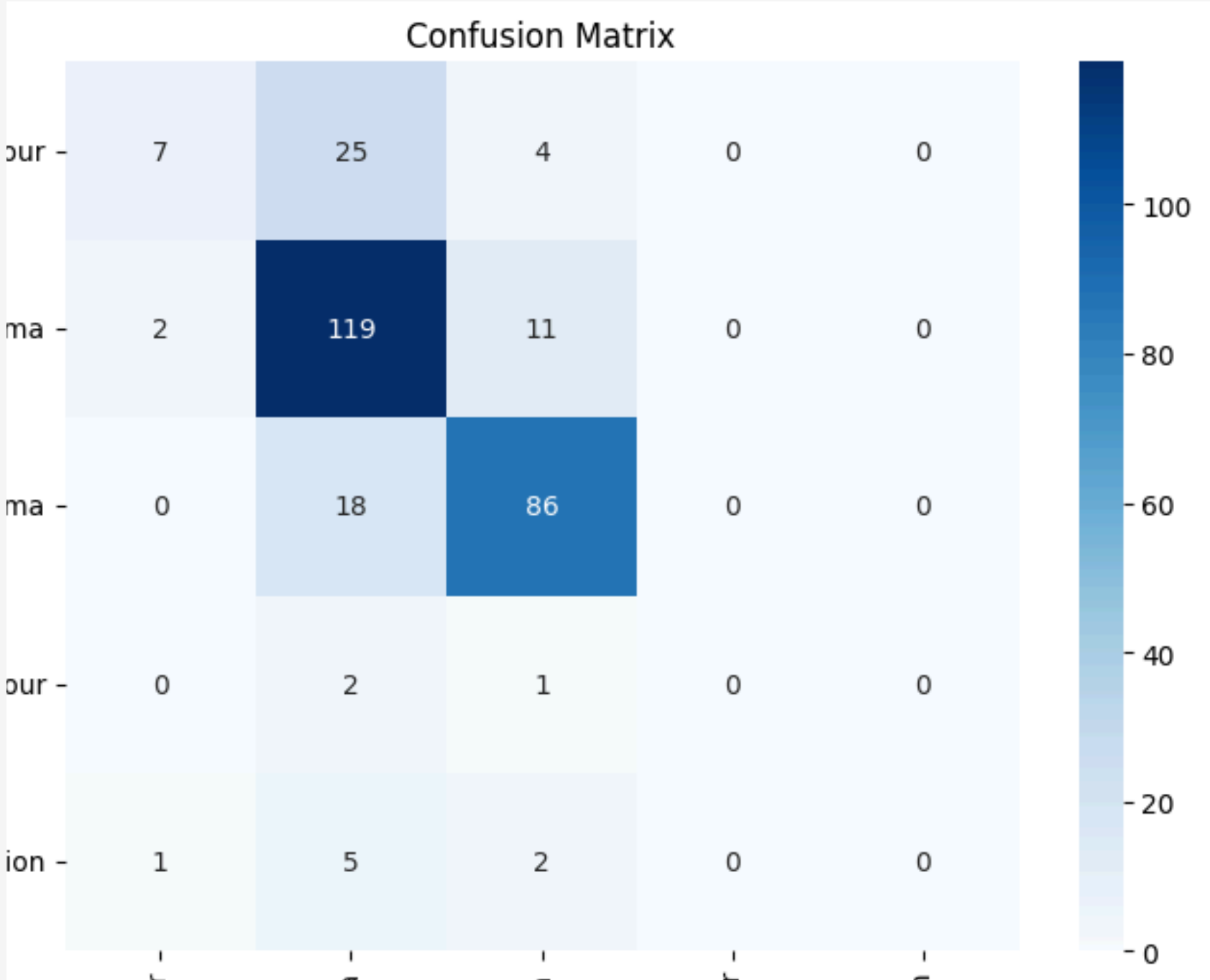
Fold 5 Micro F1: 0.7616

=====

Mean Micro F1: 0.7405

Std Micro F1: 0.0162

=====

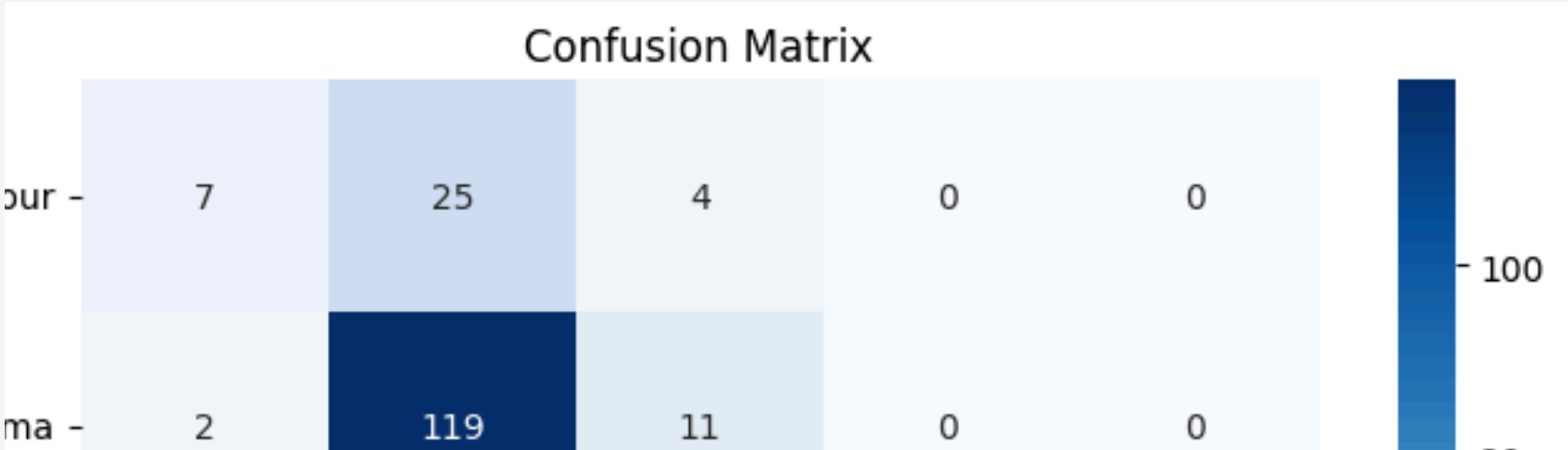


Model solely trained by Clinical Information

```
Fold 1 Micro F1: 0.7181

Fold 2
Fold 2 Micro F1: 0.7439

Fold 3
Fold 3 Micro F1: 0.7528
```





clinical_test_predictions.csv

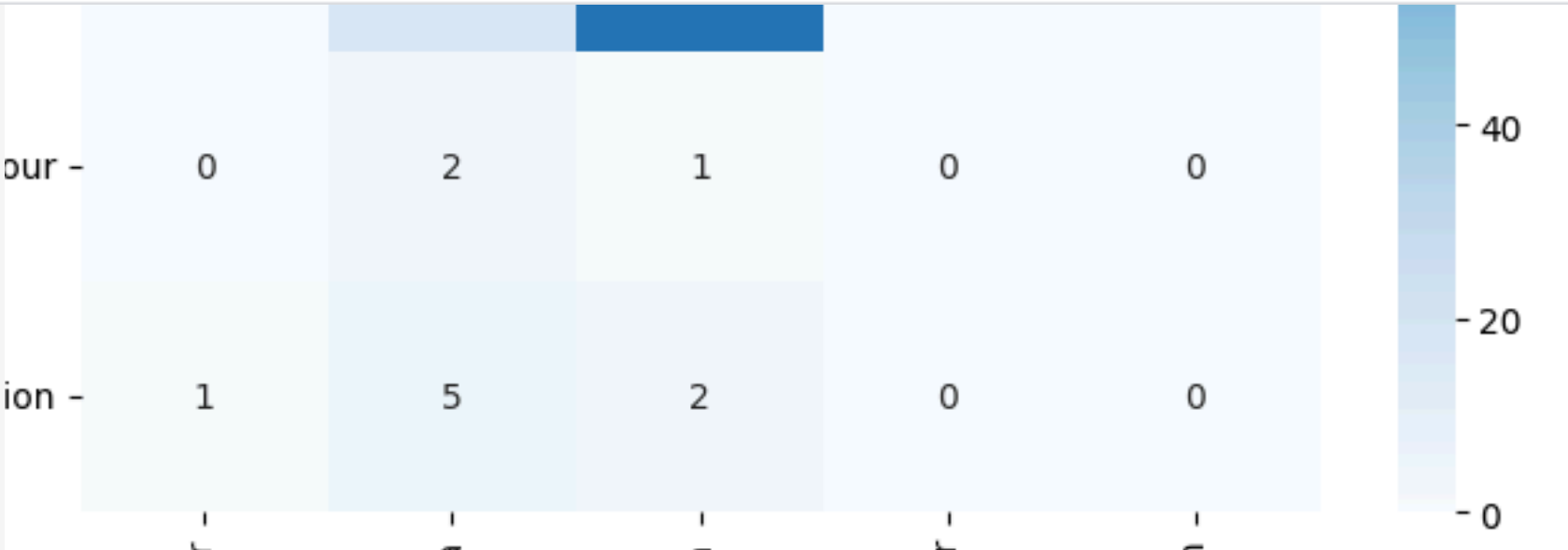
Complete · CharlesPoon · 2h ago

0.57702

```
Fold 4 Micro F1: 0.77289

Fold 5
Fold 5 Micro F1: 0.7616

=====
Mean Micro F1: 0.7405
Std Micro F1: 0.0162
=====
```



Limitations of Solely Using Clinical Features

```
Fold 1 Micro F1: 0.7181
```

```
Fold 2
```

```
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```

```
Fold 3
```

```
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```
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```

```
=====
```

```
Mean Micro F1: 0.7405
```

```
Std  Micro F1: 0.0162
```

```
=====
```

Unseen Test Data

Test set contains 163 / 378 cases (43%) with tumor locations never seen during training

Unseen Training Data

In Cross Validation Only around 18% entries have new information never seen

```
Total unique rows with ANY unseen value: 85 / 453  
Fold 2 Micro F1: 0.7439
```

Radiomics

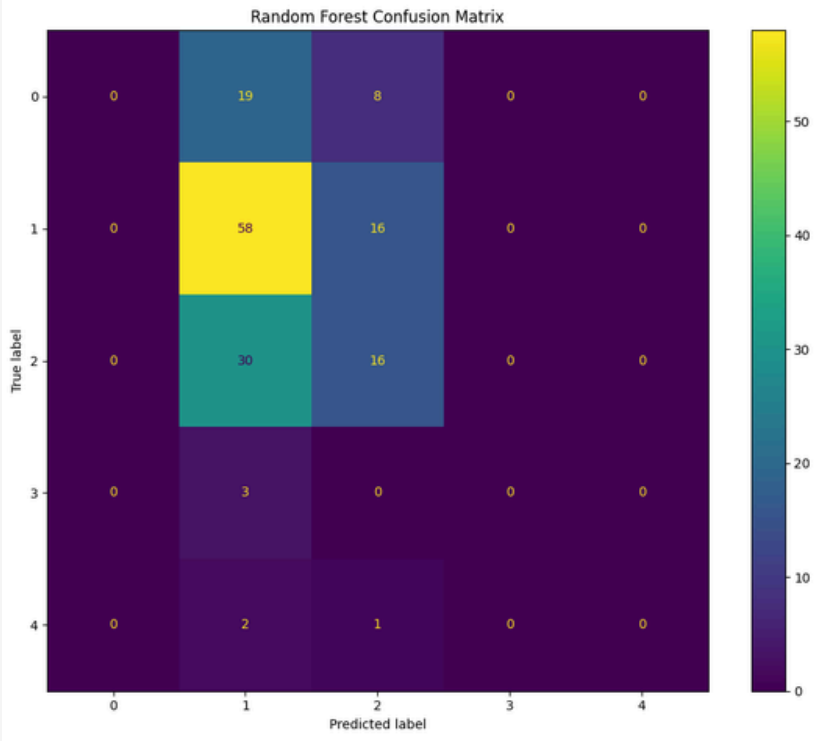
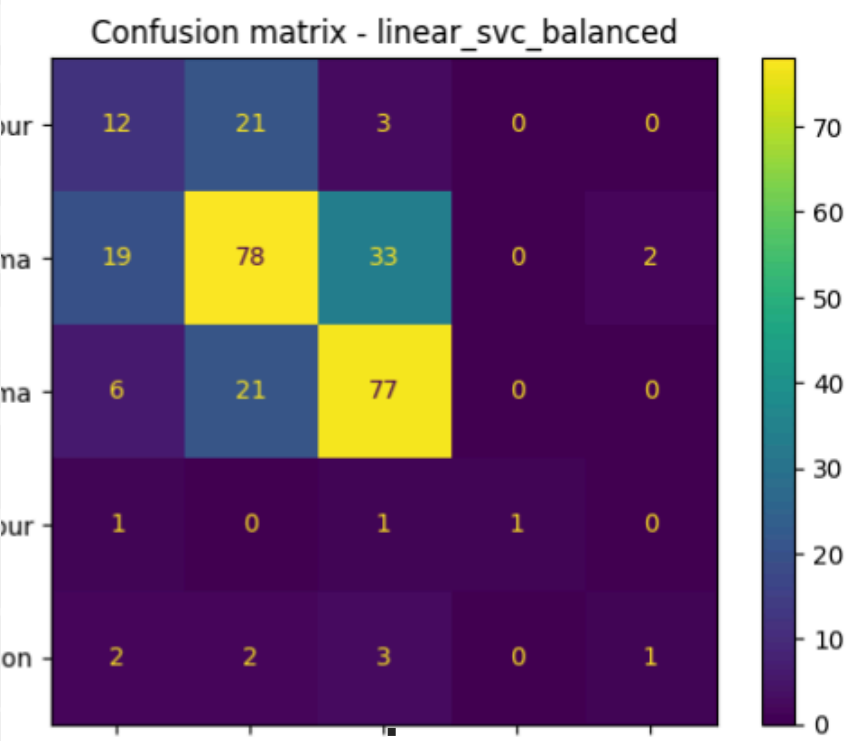
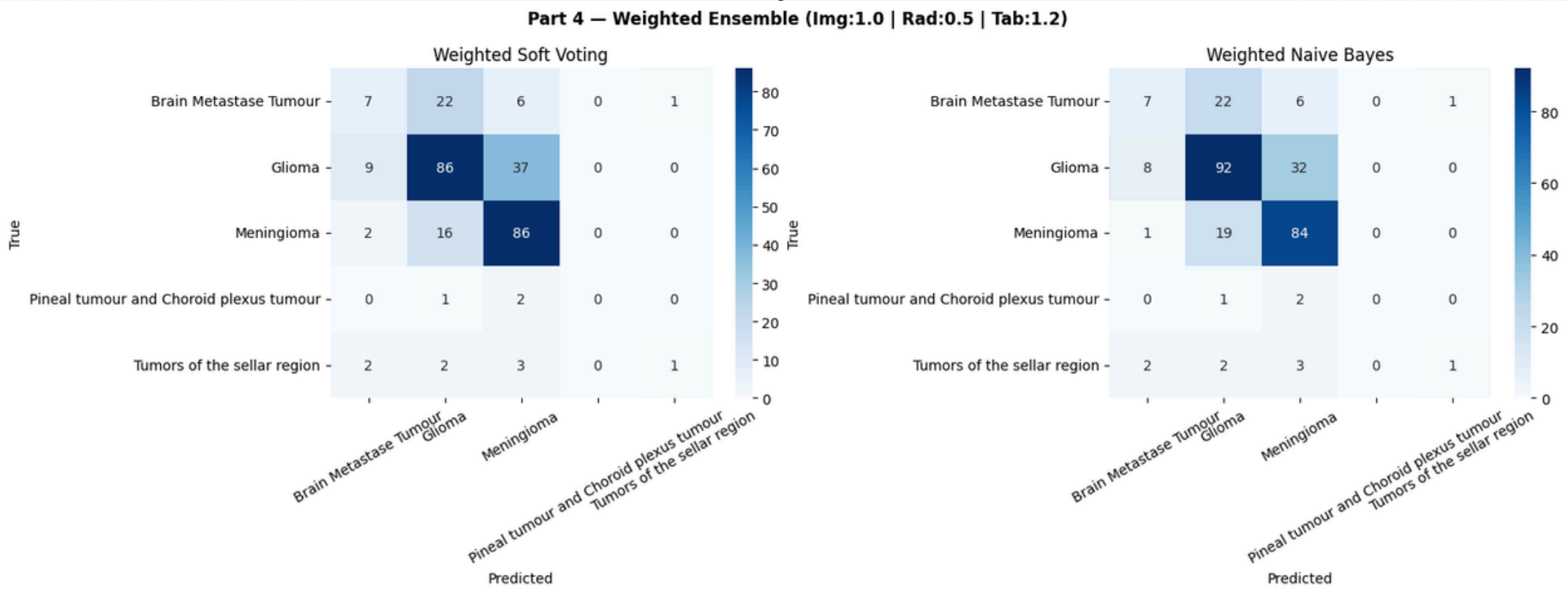
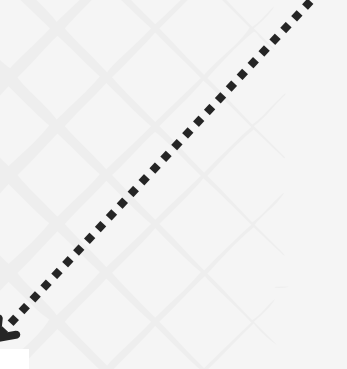
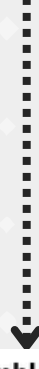
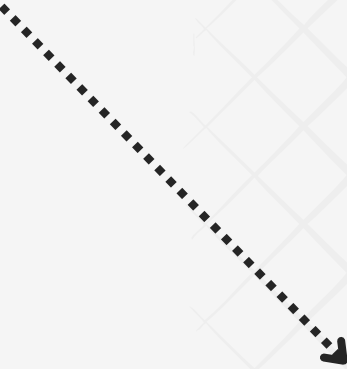
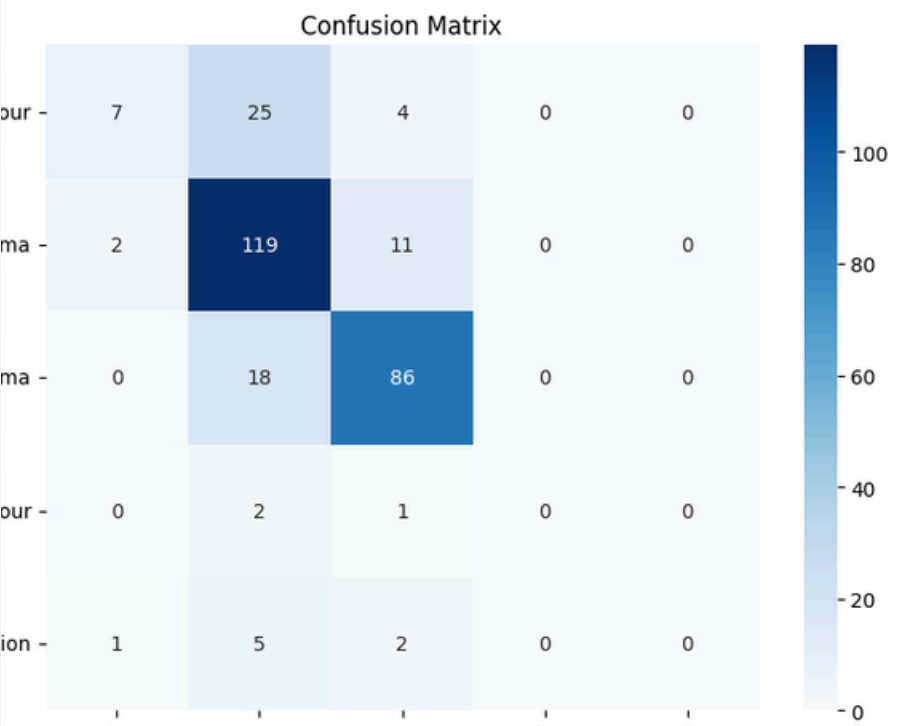


Image Features

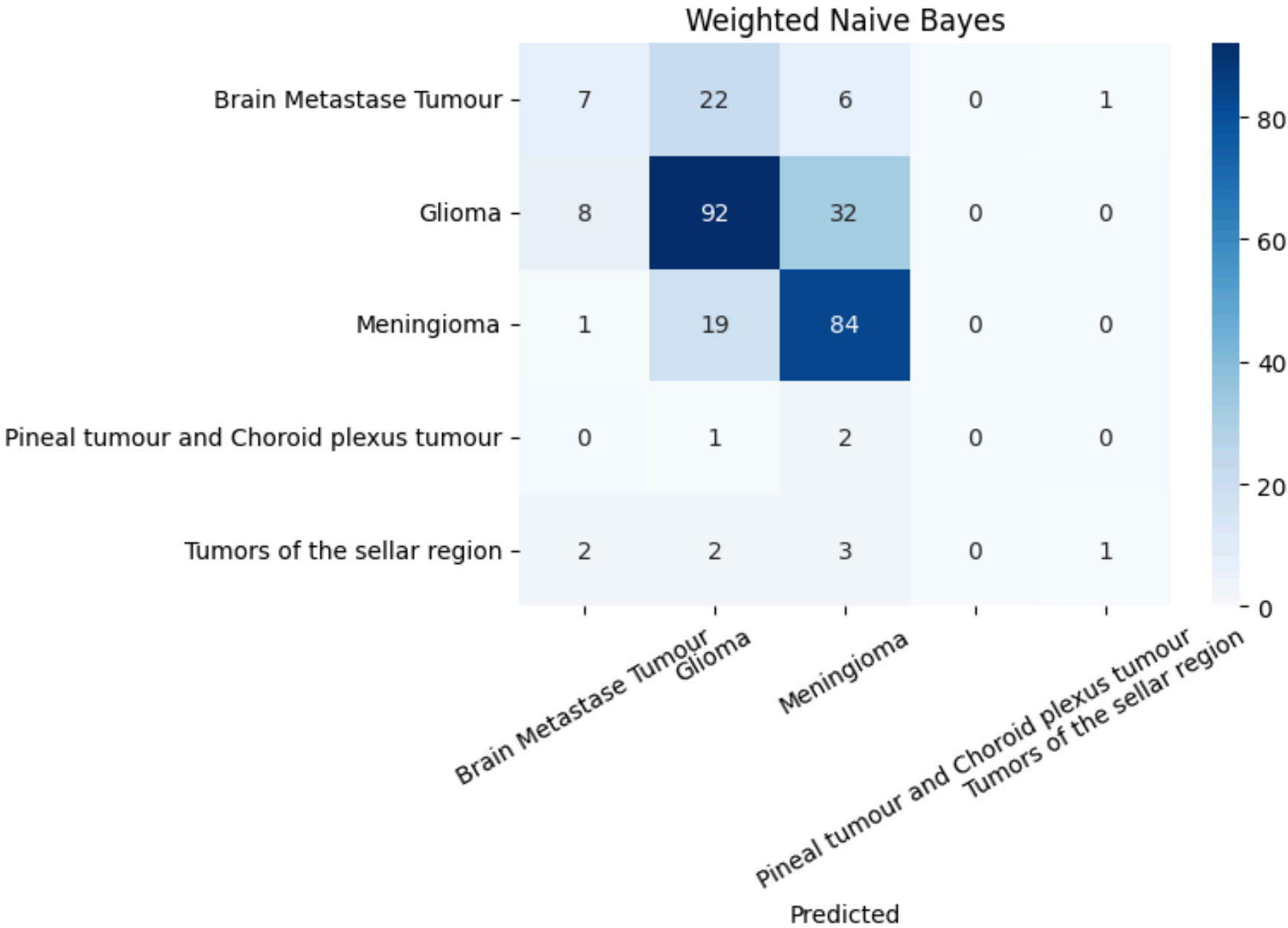
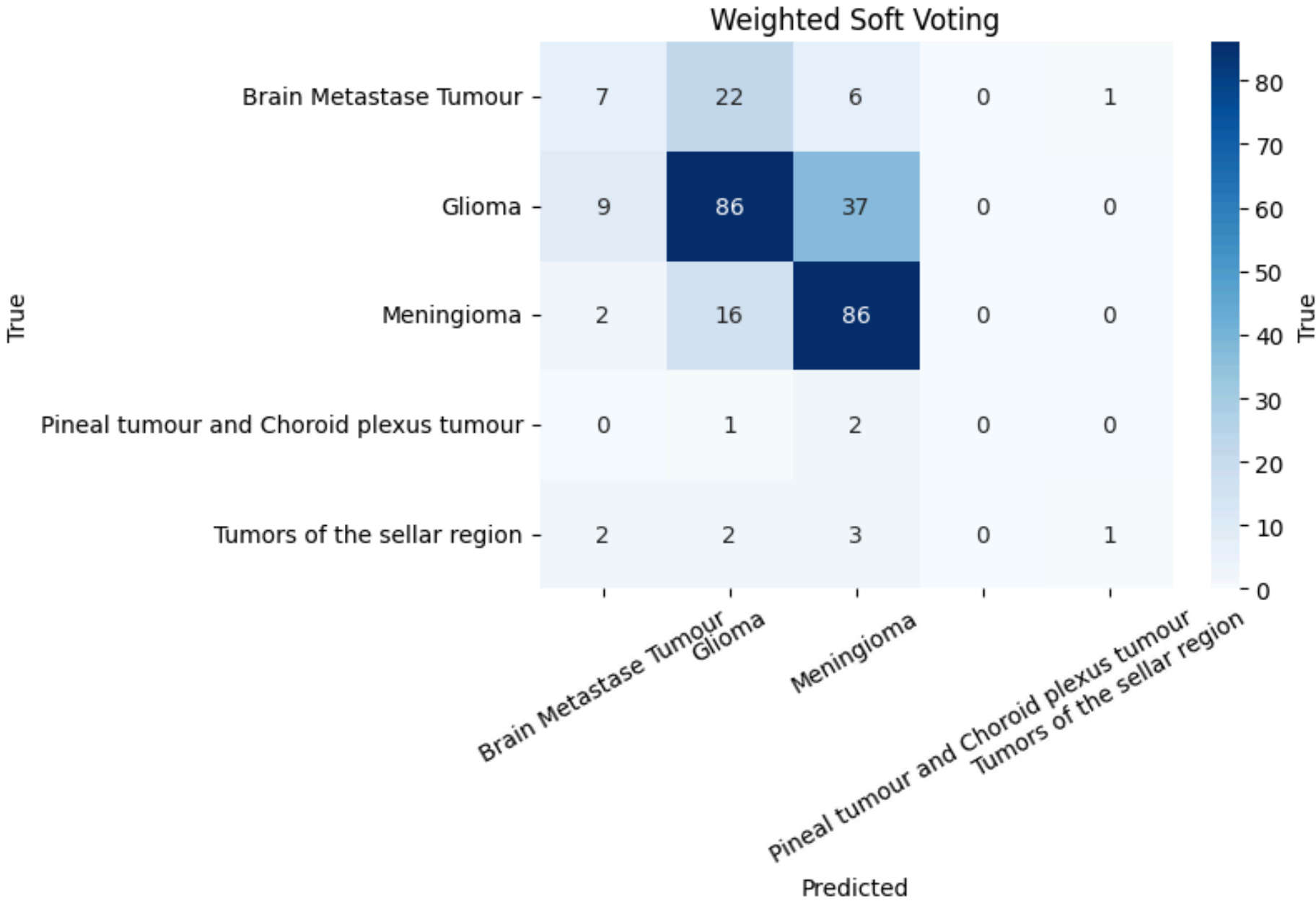


Clinical Information



Weighted Soft Voting / Naive Bayes

Part 4 — Weighted Ensemble (Img:1.0 | Rad:0.5 | Tab:1.2)



submission_final.csv

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